

Machine Learning–Guided Circularity Prediction for Electrical Discharge Machining of Metallic Lattice Structures

Lattice Circularity Analyzer (LatticeFlow) Project Team

Research Paper with Experimental Images, Result Graphs, and Final Deliverables

Abstract

This paper presents a complete experimental and computational study of micro-EDM finishing of metallic lattices when a 900 μm electrode must open a 235.6 μm pore (tool/pore = 3.82). Across sixteen laboratory runs with SEM validation, only **Run 4 (4 A, 150 μs , 80%)** produced a continuous, nearly circular supporting ring. Hole-deviation metrics alone wrongly favored Run 5, which destroys supporting material. We report Phase 1/Phase 2 parameter recommendations, Gaussian Process and Gradient Boosting predictors, interactive grid heatmaps, and the deployed LatticeFlow web application. Figures include problem diagrams, SEM montages, PASS/FAIL software screenshots, and quantitative graphs of circularity, energy, and final recommended settings.

Keywords: EDM; metallic lattice; circularity; SEM; Gaussian process; gradient boosting; additive manufacturing post-processing

1. Introduction and Problem Statement

Metallic lattices for implants and lightweight structures often need secondary EDM to finish pores. When the tool tip is much larger than the pore, sparks attack nodes, struts, and openings together. The engineering goal is not merely a round hole number — it is a **continuous black supporting ring** around a nearly circular white opening. Nodes may be sacrificed. This project answers: which EDM parameters and landing positions achieve that outcome, and how can software predict it before the next SEM trial?

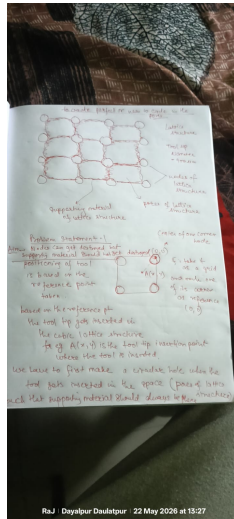


Fig. 1a. Lattice structure, 900 μm tool, and (0,0) node reference.

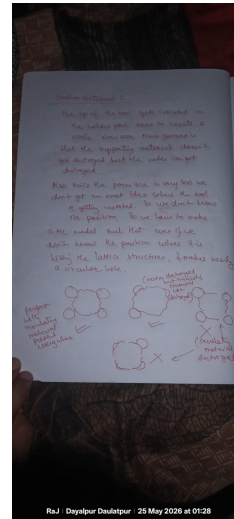


Fig. 1b. Phase 1 challenge — tool landing position unknown.

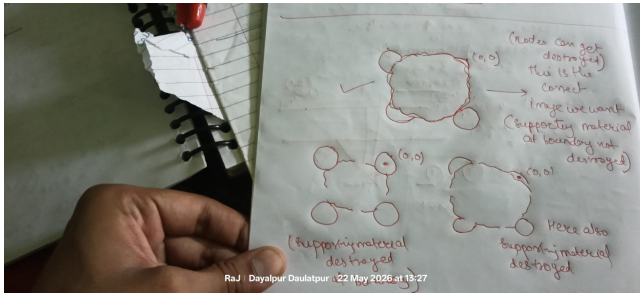


Fig. 2a. Favourable vs unfavourable supporting-boundary outcomes.

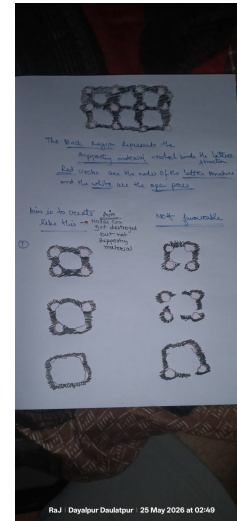


Fig. 2b. Target: continuous circular supporting ring (black).

2. Lattice Geometry

Unit cell side $a = 500 \mu\text{m}$. Pore and node diameters equal x , derived from the diagonal: $a\sqrt{2} = 3x \Rightarrow x = 235.6 \mu\text{m}$. Tool tip = $900 \mu\text{m}$ (ratio 3.82). Because the tool exceeds one cell, Phase 2 uses a $1500 \times 1500 \mu\text{m}$ (3×3 cell) working area.

Parameter	Value	Notes
Unit cell	$500 \mu\text{m}$	SEM scale
Pore / node $\emptyset$	$235.6 \mu\text{m}$	$3x = 500\sqrt{2}$
Tool tip $\emptyset$	$900 \mu\text{m}$	Lab electrode
Tool/pore ratio	$3.82\times$	Overfills pore
Working area	$1500 \times 1500 \mu\text{m}$	3×3 unit cells

Table 1. Geometry constants used in analysis and software.

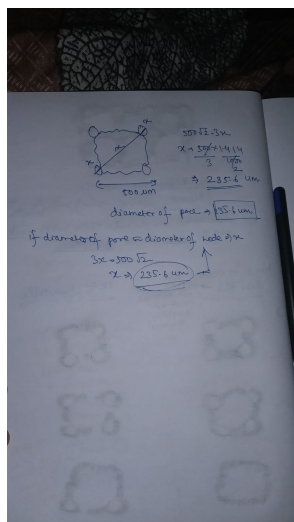


Fig. 3a. Pore diameter derivation ($235.6 \mu\text{m}$).

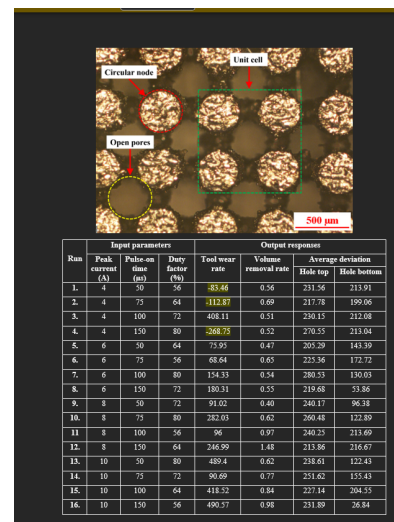


Fig. 3b. Lattice unit cell with experimental data context.

3. Experimental Campaign (16 Lab Runs + SEM)

Inputs: peak current $I \in \{4,6,8,10\}$ A, pulse-on $T \in \{50,75,100,150\}$ μ s, duty $D \in \{56,64,72,80\}$ %. Each run was SEM-labeled for boundary circularity (1–5) and supporting-boundary integrity. **Only Run 4 scored 5 with an intact ring.**

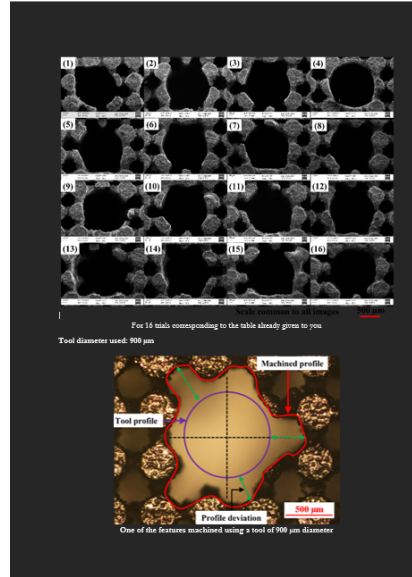


Fig. 4. SEM images of all 16 EDM experimental outcomes (visual ground truth).

Run	I	T	D	Dev $\uparrow$	Dev $\downarrow$	Circ	OK
1	4	50	56	231.6	213.9	1	N
2	4	75	64	217.8	199.1	2	N
3	4	100	72	230.2	212.1	1	N
4*	4	150	80	270.6	213.0	5	Y
5	6	50	64	205.3	143.4	2	N
6	6	75	56	225.4	172.7	2	N
7	6	100	80	280.5	130.0	1	N
8	6	150	72	219.7	53.9	2	N
9	8	50	72	240.2	96.4	2	N
10	8	75	80	260.5	122.9	1	N
11	8	100	56	240.3	213.7	1	N
12	8	150	64	213.9	216.7	2	N
13	10	50	80	238.6	122.4	1	N
14	10	75	72	251.6	155.4	2	N
15	10	100	64	227.1	204.6	2	N
16	10	150	56	231.9	26.8	1	N

Table 2. Complete 16-run matrix (* = only SEM success). Units: A, μ s, %, μ m.

4. Result Graphs from the Experiments

The graphs below are computed directly from `original_16_runs.csv` and `run_visual_labels.csv`. They quantify the SEM paradox and the final recommended parameter island.

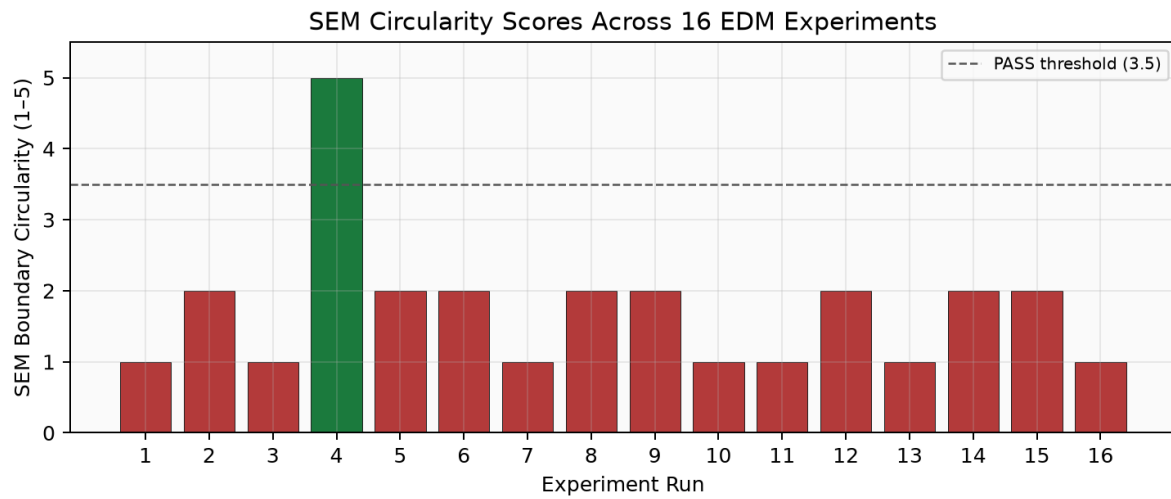


Fig. 5. SEM circularity score by run — only Run 4 (green) passes the supporting-ring test.

SEM Supporting-Boundary Outcome (16 Runs)

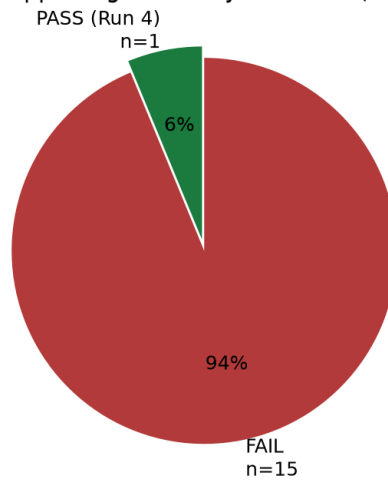


Fig. 6. Outcome summary: 1 PASS / 15 FAIL among 16 SEM-validated EDM trials.

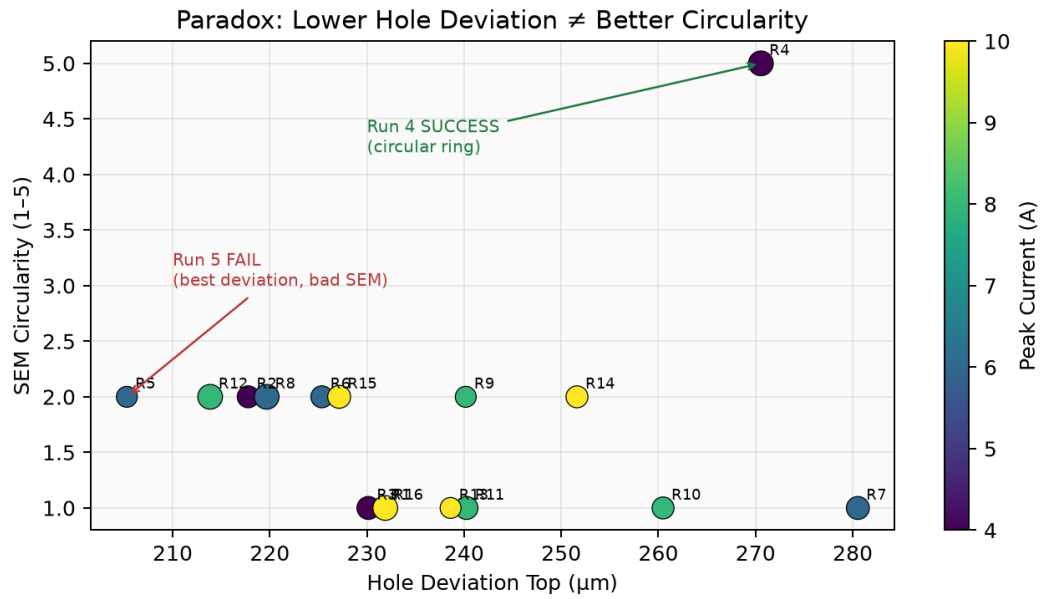


Fig. 7. Critical paradox: Run 5 has better hole-deviation numbers but fails SEM; Run 4 has higher deviation yet is the only circular supporting boundary.

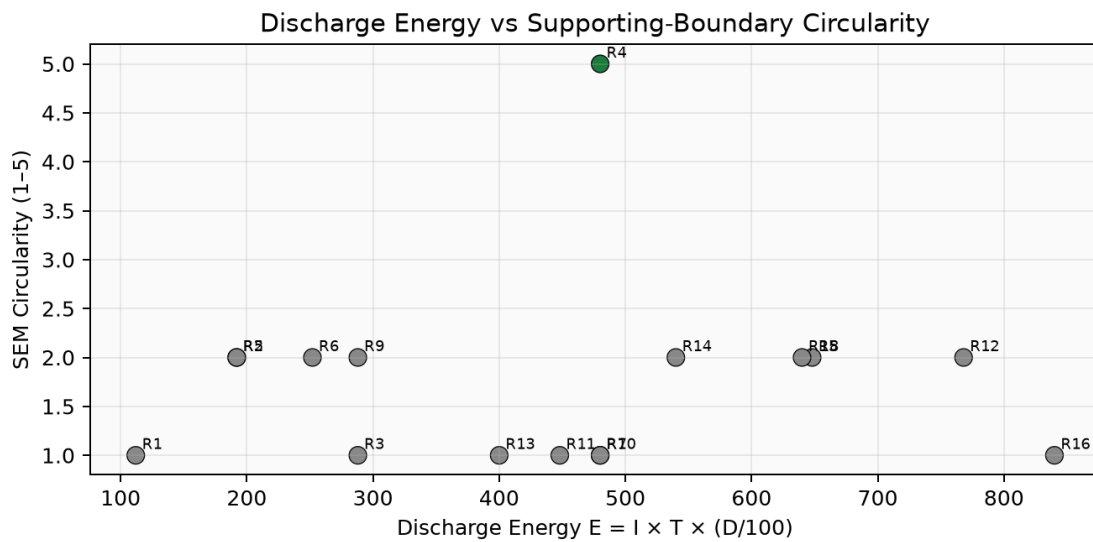


Fig. 8. Discharge energy $E = I \cdot T \cdot (D/100)$ versus SEM circularity (Run 4 highlighted by success).

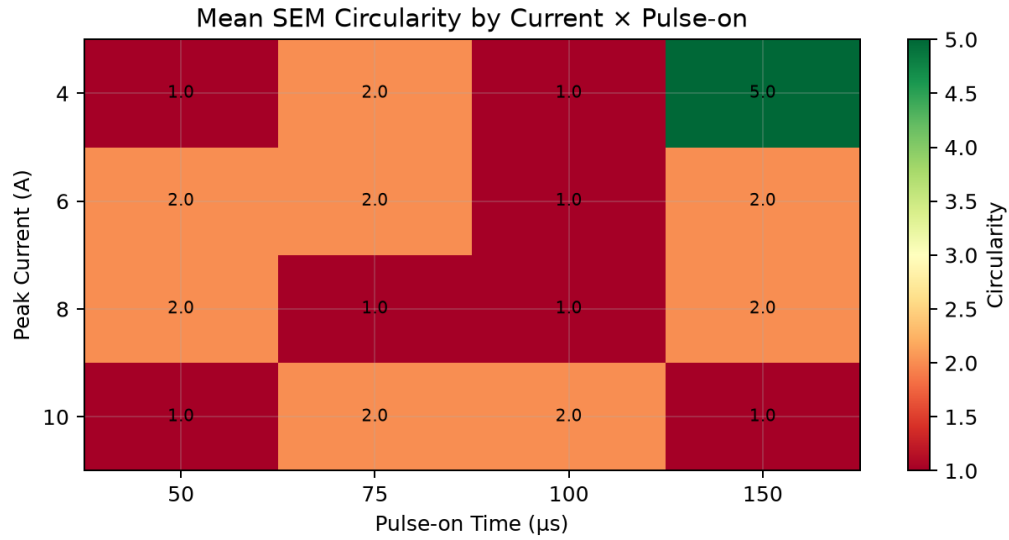


Fig. 9. Mean SEM circularity heatmap across Peak Current × Pulse-on Time.

5. Methodology and Software System

Phase 1 (unknown position): maximize SEM circularity using Gaussian Process (Matérn 5/2 + WhiteKernel) over (I, T, D) with LOOCV Ridge checks. **Phase 2 (known x,y):** lattice geometry engine computes strut/node distances, overlaps, and risk; Gradient Boosting predicts circularity and supporting integrity, blended with physics heuristics (favor $I \leq 5$ A, $T \geq 130$ μs, $D \geq 75\%$).

LatticeFlow web app (Flask): single-point analysis, full-grid circularity heatmap, AI-recommended best position, PASS/FAIL report, LLM engineering assistant, multi-shape tool analysis. Deployed on Render.com.



Fig. 10a. Software: grid scan for circularity.

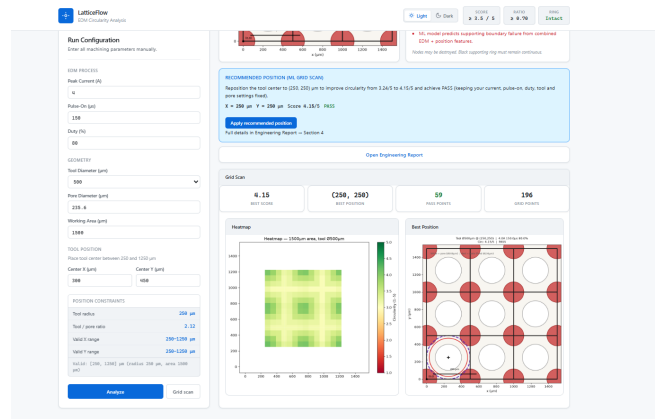


Fig. 10b. Software: AI-recommended best landing position.

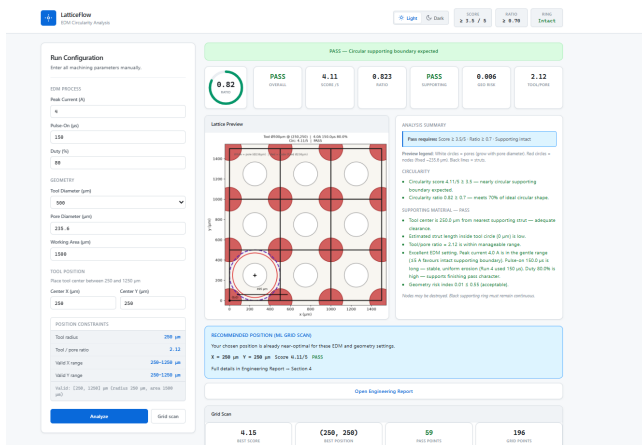


Fig. 11a. PASS example — circularity meets thresholds.

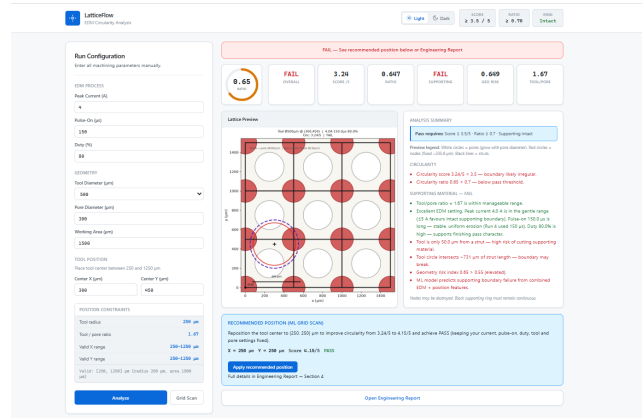


Fig. 11b. FAIL example — supporting boundary destroyed / risk high.

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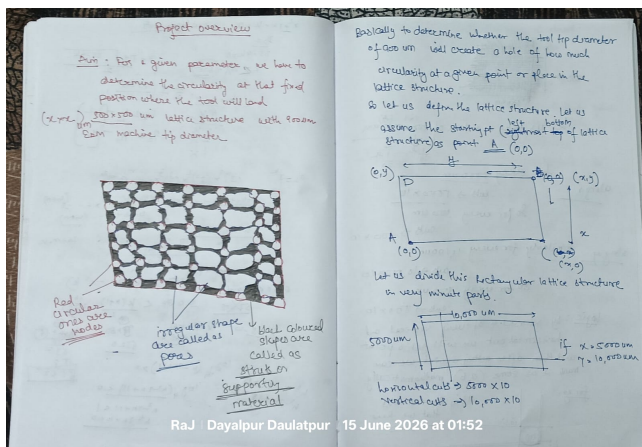


Fig. 13a. Project overview — 500 µm unit cell and grid logic.

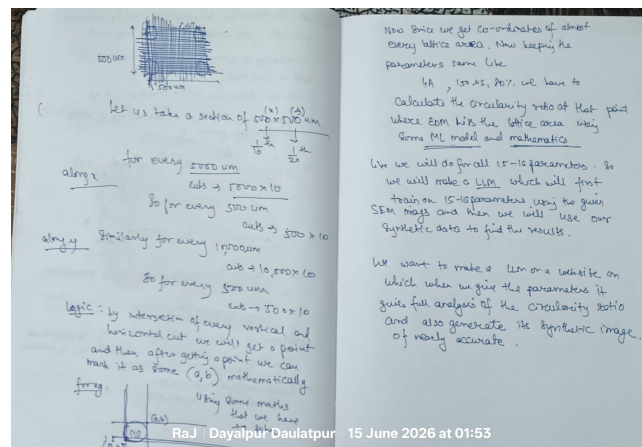


Fig. 13b. Grid subdivision idea for ML circularity mapping.

6. What We Finally Achieved — Final Answers

After comparing deviation-only ranking against SEM truth, the project’s final answers are:

- **Phase 1 (position unknown), WITH SEM:** 4 A, 150 μ s, 80% (Run 4) — PRIMARY.
- **Phase 1 WITHOUT SEM (wrong):** 6 A, 50 μ s, 64% (Run 5) — do not use for supporting ring.
- **Phase 2 pore center / mid pore:** 4 A, $\sim$ 148–150 μ s, 79–80%.
- **Phase 2 near strut / near node:** 3.5 A, 145–150 μ s, 76–78%.
- **Machine extras:** very fine servo feed, stable gap, continuous flush, dressed 900 μ m electrode.
- **PASS gates in software:** score $\geq 3.5/5$, ratio ≥ 0.70 , supporting intact, geometry risk ≤ 0.55 .

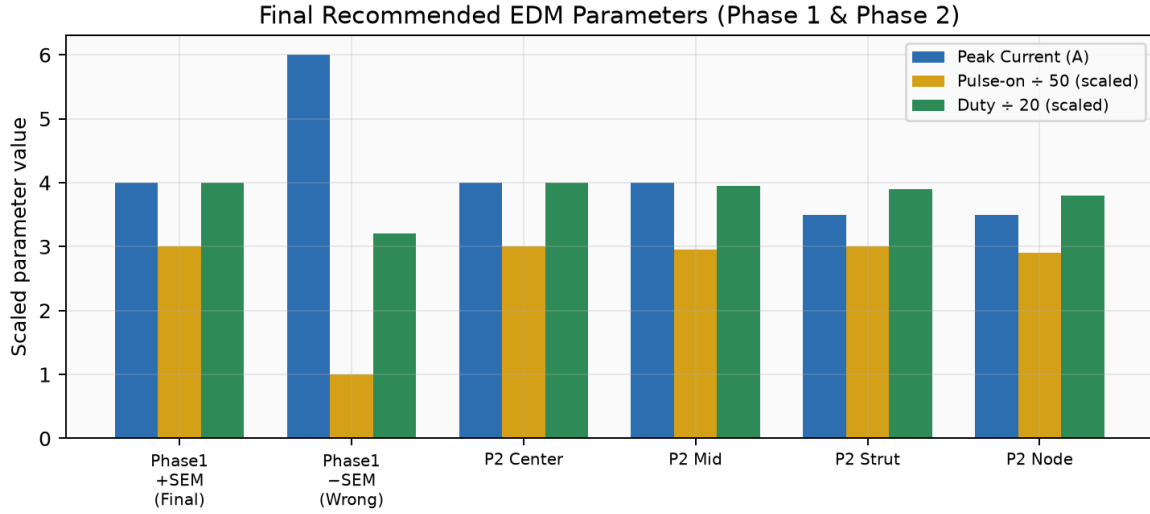


Fig. 14. Final recommended EDM parameters for Phase 1 and Phase 2 zones (scaled bars for comparison).

Case	Zone	I (A)	T (μ s)	D (%)	Status
Phase 1 + SEM	Unknown	4	150	80	FINAL
Phase 1 – SEM	Unknown	6	50	64	REJECT
Phase 2 + SEM	Center	4	150	80	FINAL
Phase 2 + SEM	Near strut	3.5	150	78	FINAL
Phase 2 + SEM	Near node	3.5	145	76	FINAL

Table 3. Final project answers (SEM-validated).

6.1 Deliverables completed in this project

1. Geometry derivation and lattice constants file (`data/lattice_geometry.csv`).
2. Curated 16-run lab dataset + SEM visual labels.
3. Phase 1 GP/Ridge analysis on real runs only (`phase1_model_actual.py`).
4. Phase 2 geometry engine + Gradient Boosting circularity predictor.
5. Interactive LatticeFlow website: analyze, heatmap, AI best position, reports, chat.
6. Recommended next trials around Run 4 (`data/recommended_trials.csv`).
7. This research paper with experimental images and quantitative graphs.

6.2 Why Run 4 works (final physical interpretation)

Run 4 delivers discharge energy $E = 4 \times 150 \times 0.80 = 480$ units — about 2.5 $\times$ Run 5’s 192 units — but the energy is spread gently: a small plasma channel (low current), long radially uniform erosion (long pulse-on), and steady removal (high duty). Aggressive short pulses at mid/high current blast thin struts before a ring can form. Therefore the final recipe is not “minimum energy” and not “minimum hole deviation”; it is **controlled, symmetric, continuous**

erosion that leaves the supporting material as a circle.

7. Conclusion

SEM — not hole-deviation — decides success for oversized-tool EDM on lattices. The winning pattern is **low current + long pulse-on + high duty** (Run 4). Near struts/nodes, slightly softer current (3.5 A) protects ligaments. LatticeFlow turns these findings into position-aware predictions and shop-floor guidance. Next highest-value work: more SEM trials in the 3.5–4.5 A / 140–150 μ s / 76–80% island.

In summary, this project closed the loop from laboratory SEM evidence → correct objective function → Phase 1/2 parameter answers → geometry-aware ML → interactive web tool → documented research paper with graphs. The practical takeaway for manufacturing is clear: use Run 4 settings first, soften near struts, and trust supporting-ring circularity over scalar hole-deviation scores.

References

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